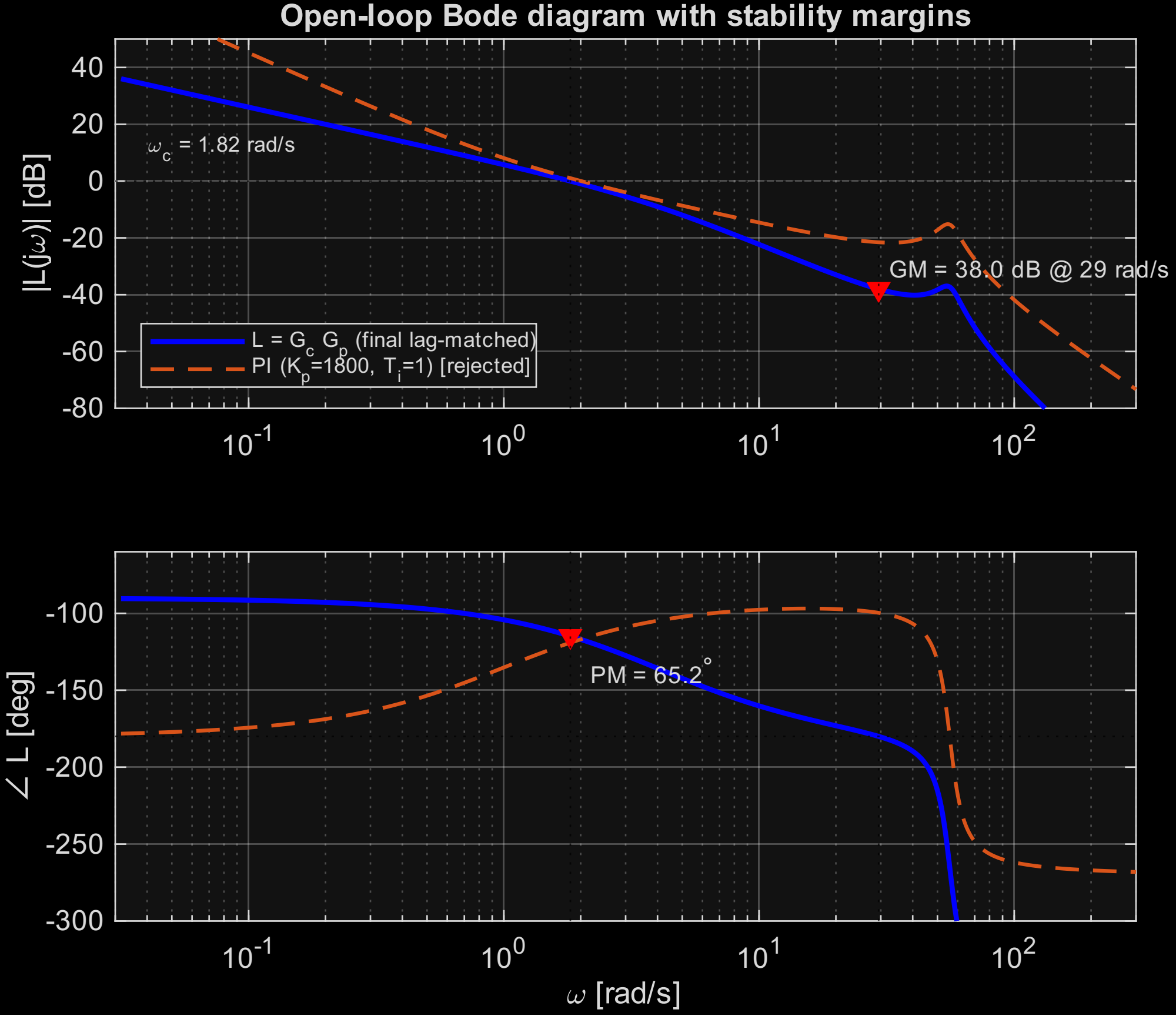

Open-loop Bode diagram with stability margins
|L(jω)| [dB]
40
20
0
-20
-40
-60
-80
ω_c = 1.82 rad/s
GM = 38.0 dB @ 29 rad/s
L = G_c G_p (final lag-matched)
PI (K_p=1800, T_i=1) [rejected]
10^-1
10^0
10^1
10^2
∠ L [deg]
-100
-150
-200
-250
-300
PM = 65.2°
ω [rad/s]